Zeyuan Yang

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EDUCATION

University of Massachusetts, Amherst, USA . . . . . *Aug 2024 - Present*
Ph.D. student in Computer Science
Advisors: Chuang Gan, Hao Zhang
Research Interests: Embodied Intelligence, Multi-modal Foundation Models, World Models

Tsinghua University, Beijing, China . . . . . *Sep 2021 - July 2024*
M.S. in Computer Science
Advisor: Yang Liu
Research Interests: Natural Language Processing, Continual Learning

Tsinghua University, Beijing, China . . . . . *Sep 2017 - July 2021*
B.S. in Information Management, minor in Data Science
Major GPA: 3.63/4.0 (Rank 4/29)
Thesis Advisor: Qing Ye

PUBLICATIONS (Google Scholar, * denotes equal contribution.)

[13] Perception or Reasoning? Diagnosing and Tackling Reinforcement Learning Bottlenecks in VLMs
Zeyuan Yang*, Zhuohao Chen*, Jiaben Chen, Yao Zhang, Maohao Shen, Chuang Gan
In submission, 2026

[12] Multimodal Thinking with Renderable Programs
Sunli Chen, Ding Zhong, Ziqiao Ma, Jiaxin Liu, **Zeyuan Yang**, Hao Zhang, Joyce Chai, Chuang Gan
In submission, 2026

[11] UniCanvas: A Diffusion-based Unified Model with Text-in-Image Joint Generation
Zeyuan Yang*, Hao-Wei Chen*, Yuncong Yang, Xueyang Yu, Haoyu Zhen, Ziqiao Ma, Maohao Shen, Chuang Gan
Best Paper at *CVPR A2A-MML Workshop*, 2026
In submission, 2026

[10] SyncWorld: Visual Calibration Enables World Models as Zero-Shot Simulators
Yuncong Yang, Zhengtao Han, Furkan Ozyurt, **Zeyuan Yang**, Han Yang, Junyi Cao, Haoyu Zhen, Yilun Du, Chuang Gan
NeurIPS Oral, 2026

[9] Machine Mental Imagery: Empower Multimodal Reasoning with Latent Visual Tokens
Zeyuan Yang*, Xueyang Yu*, Delin Chen, Maohao Shen, Chuang Gan
Best Paper at *ICCV KnowledgeMR Workshop*, 2025
CVPR, 2026

[8] VCA: Video Curious Agent for Long Video Understanding
Zeyuan Yang*, Delin Chen*, Xueyang Yu, Maohao Shen, Chuang Gan
ICCV, 2025

[7] UBSoft: A Simulation Platform for Robotic Skill Learning in Unbounded Soft Environments
Chunru Lin, Jugang Fan, Yian Wang, **Zeyuan Yang**, Zhehuan Chen, Lixing Fang, Tsun-Hsuan Wang, Zhou

Xian, Chuang Gan
CoRL, 2024

[6] Rethinking Long Context Generation from the Continual Learning Perspective
Zeyuan Yang, Fangzhou Xiong, Peng Li and Yang Liu
COLING, 2025

[5] Towards Unified Alignment Between Agents, Humans, and Environment
Zonghan Yang, An Liu, Zijun Liu, Kaiming Liu, Fangzhou Xiong, Yile Wang, **Zeyuan Yang**, Qingyuan Hu, Xinrui Chen, Zhenhe Zhang, Fuwen Luo, Zhicheng Guo, Peng Li, Yang Liu
ICML, 2024

[4] Reflective and Imaginative Language Agent for Zero-Shot Semantic Audio-Visual Navigation
Zeyuan Yang*, Jiageng Liu*, Peihao Chen, Anoop Cherian, Tim Marks, Jonathan Le Roux, Chuang Gan
CVPR, 2024

[3] Failures Pave the Way: Enhancing Large Language Models through Tuning-free Rule Accumulation
Zeyuan Yang, Peng Li, Yang Liu
EMNLP, 2023

[2] Restricted Orthogonal Gradient Projection for Continual Learning
Zeyuan Yang, Zonghan Yang, Yichen Liu, Peng Li, Yang Liu
AI Open, 2023

[1] Dynamic Multi-Branch Layers for On-Device Neural Machine Translation
Zhixing Tan, **Zeyuan Yang**, Meng Zhang, Qun Liu, Maosong Sun, Yang Liu
IEEE/ACM Transactions on Audio, Speech, and Language Processing, 2022

[0] Exploring deep learning for air pollutant emission estimation
Lin Huang, Song Liu, **Zeyuan Yang**, Jia Xing, Jia Zhang, Jiang Bian, Siwei Li, Shovan Kumar Sahu, Shuxiao Wang, Tie-Yan Liu
Geoscientific Model Development , 2021

EMPLOYMENT

Amazon's Frontier AI & Robotics . . . . . *May 2026 - Now*
Research Intern, Supervisors: Yunsheng Tian, Chen Feng, Rocky Duan — San Francisco, USA
Navigation Foundation Agents

Samsung Research America . . . . . *May 2025 - Aug 2025*
Research Intern, Supervisors: Luowei Zhou, Deen Dayal Mohan — Mountain View, USA
Multi-modal Retrieval and Foundation Models

Zhiyuan Innovation Technology Co., Ltd. . . . . . *Jan 2024 - Aug 2024*
Research Intern, Agibot Lab, Supervisor: Hao Dong — Beijing, China
Humanoid Robotics

TEACHING EXPERIENCE

Information Systems, University of Massachusetts, Amherst . . . . . Spring 2026
Teaching Assistant, Instructor: Trek Palmer

Foundations of Programming, University of Massachusetts, Amherst . . . . . Spring 2025
Teaching Assistant, Instructor: Cole Adam Reilly

Objected Oriented Programming, University of Massachusetts, Amherst . . . . . Fall 2024
Teaching Assistant, Instructor: Jaime Davila